

Summarizing SEC 10-K MDA with Long-Context Transformers: KPI-Aware Evaluation for Financial Narratives

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Abstract—Key performance indicators (KPIs) that are important to analysts are frequently overlooked by generic summarizers in the lengthy, jargon-heavy financial MD&A sections. This paper investigates the use of long-context transformers for KPI-aware abstractive summarization of SEC 10-K MD&A. By aligning the Summarized 10K-MDA dataset with the Financial-Reports-SEC corpus, we create a joint benchmark that produces full-length MD&A documents along with document-level KPI sets and human summaries. We refine long-context models (LED, BigBird-Pegasus, LongT5) and short-context baselines (BART, PEGASUS) on this benchmark using a single pipeline. We then assess using ROUGE, BLEU, semantic similarity, compression ratio, and a list of matched, hallucinated, and missing KPIs. A tuned BART variant achieves the best lexical and semantic scores, according to experiments on a 153-document test set, while long-context models provide competitive but not obviously better results. The difference between surface summary quality and financial usefulness is highlighted by the fact that KPI-level F1 is consistently low across all architectures and that the majority of errors result from missing rather than fake KPIs.

Index Terms—Abstractive summarization, financial narrative, long-context transformers, KPI fidelity, BART, PEGASUS, BigBird, LongT5, ROUGE, chrF

I. INTRODUCTION

Organizations, whether small or big, generate various reports during their financial year, containing both narrative and quantitative information, such as annual financial reports. The objective of these reports is to present the organization's operating results, financial position, and cash flows [1]. Companies rely on financial narrative to engage with stakeholders including investors, shareholders, customers, employees, financial analysts, regulators, lenders, rating agencies, and suppliers. It allows stakeholders to assess how well a company is performing and creating value [2].

The advent of artificial intelligence and machine learning has transformed the way financial data is processed and analyzed. With an increasing amount of data available, there is a need for effective summarization techniques that can extract essential insights while preserving the context of complex financial reports. Although advancements in financial AI have transformed how reports are generated and analyzed, annual reports and MD&A sections continue to be lengthy, dense

with numerical data, and heavily reliant on domain-specific terminology. Generic abstractive systems often truncate inputs or paraphrase loosely, risking omissions or distortions of key performance indicators (KPIs) such as revenue, EBITDA, margin, and VAT.

This study targets financial narrative summarization on the Summarizing SEC 10-K MDA using long context transformers that can ingest thousands of tokens in one pass. We compare strong short-context baselines (BART, PEGASUS) with long-context models (e.g., BigBird-Pegasus, LongT5), and evaluate not only with ROUGE/chrF but also with a lightweight KPI/terminology checklist that directly measures domain fidelity. As an optional ablation, we probe whether adding simple table cues improves numeric coverage without increasing system complexity. The goal is a practical, reproducible pipeline that produces concise, trustworthy summaries for accounting workflows.

II. LITERATURE REVIEW

A review of recent research (2020–2025) highlights the rapid evolution of Transformer-based architectures for long document summarization and understanding. Early work identified that conventional Transformers struggle with lengthy sequences due to fixed token limits and quadratic attention complexity. To address this, Zaheer et al. introduced BigBird [8], which integrates sparse attention combining local, global, and random patterns, enabling efficient processing of sequences up to eight times longer than standard Transformers. Khanna et al. [3] conduct an extractive summarization of very long annual financial reports in the FNS 2022 shared task using the LED model, which ingests up to 8,192 input tokens and 1,024 target tokens. The authors investigate sequential (intermediate) fine-tuning on the arXiv summarization dataset before FNS but observe only marginal gains, while still outperforming prior FNS baselines on ROUGE metrics. The paper points out that detecting the full summary span in a long document remains a challenge and suggests chunking and graph-based models as potential future work. The Longformer-Encoder-Decoder (LED) model [6] was applied to financial document summarization, revealing that direct

fine-tuning is often more effective than sequential methods, though performance remains constrained by limited domain datasets and metric inadequacies such as ROUGE’s inability to capture factuality. Complementary to this, USTT [7] addressed multimodal summarization by combining textual and tabular data, emphasizing the importance of hybrid architectures for real-world applications. Rizki et al. [4] develop an LLM-based auto-summarization system to enable Indonesian retail investors to better digest long corporate annual reports. They build a custom dataset of 2,008 narrative-summary pairs extracted from 424 IDX-listed companies and fine-tune Llama-3.2-3B-Instruct using LoRA (PEFT) with a financial analyst-style prompt. The fine-tuned model attains significant ROUGE gains over the zero-shot baseline-e.g., +18.63% ROUGE-1, +44.45% ROUGE-2, +33.83% ROUGE-L-and coherently generated summaries capturing key KPIs and macro context. They further wrap the model in a Gradio web prototype that can parse PDFs, apply a map-reduce summarization pipeline, and output investor-focused report overviews. Hu et al.’s SSP framework [5] further contributed a post-editing approach that prunes structural errors using dependency-tree analysis, complementing earlier entity-level correction methods.

III. METHODOLOGY

A. Data Sources

1) *Summarized_10K-M&DA*: The first dataset used in this work is the summarized 10k-MDA collection from hugging-face. It contains management discussion sections from U.S. 10-K filings paired with abstractive summaries and automatically extracted financial metrics. The initial summaries were produced using a LongT5 and then pos-processed; the dataset provides metadata such as CIK, Company name, filing data, and a compact summary for MD&A segment.

For this study, we selected a reduced evaluation sample of 153 firm- year MD&A segment to ensure that fine- tuning procedures and manual analysis remained manageable with in a single- GPU computational setting. From this dataset we use:

- the MD&A segment text (as the “reference” document in the KPI dataset)
- the abstractive summary (as the gold summary)
- the list of extracted KPIs associated with each document

This dataset provides high-quality labels (summaries+KPI sets), but the context length remains constrained, as the MD&A sections are truncated around the LongT5 window in the original work.

2) *Financial-Reports-SEC*: The second dataset, the financial reports-sec dataset, contains sentence-level segments that originate from SEC filing documents that include complete 10-K reports with detailed metadata, including CIK numbers and filing dates, and section ID labels that pinpoint item 7 (MD&A) and other sections. The dataset allows us to create complete MD&A sections through sentence aggregation based on their MD&A section label, which match “section 7” and other similar labels.

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label, which match “section 7” and other similar labels. This gives us document lengths in the order of several thousand tokens, which is necessary to exercise long-context transformer architectures such as LED, LongT5, and BigBird Pegasus.

3) *Joint Usage*: The two datasets are integrated by matching using the CIK and filing date. For each record in the summarized 10-K MD&A dataset, we locate the corresponding filing in the financial reports SEC collection and attach the fully reconstructed MD&A section. As a result, each final sample contains the following components:

- Source document: the full MD&A section from financial-reports-sec.
- Reference summary: the abstractive summary from Summarized_10K-MDA.
- Document KPIs: KPI set derived from Summarized_10K-MDA.

This integration enables the training and assessment of long-context summarization models on MD&A sections of realistic length, while still using the compact, curated summaries and KPI annotations provided in the summarized 10-K MD&A dataset.

B. Data Preprocessing

Our preprocessing pipeline has three goals: (i) reconstruct clean MD&A text, (ii) normalize KPIs, and (iii) make the data compatible with various transformer tokenizers.

1) *MD&A extraction and cleaning*: From the financial reports- SEC dataset, all sentences are grouped by (CIK, Filing date, section ID), and those corresponding to the MD&A section are concatenated to form a continuous document. During this process, we remove EDGAR headers, boilerplate page markers, and residual HTML artifacts, followed by whitespace normalization and the removal of duplicated lines. Extremely short or incomplete segments labeled as MD&A, typically arising from parsing inconsistencies, are discarded to maintain data quality.

2) *KPI normalization*: Summarized_10K-MDA provides financial metrics as key–value pairs. We normalize KPI names to a small controlled vocabulary (e.g., revenue, net income, eps, operating income), lowercase everything, and strip punctuation from numeric fields so that simple string matching suffices for most KPI mentions. The normalized dataset serves as the document-level KPI reference standard.

3) *Summary and document normalization*: Summaries and MD&A text are lowercased, excess newlines are removed, and Unicode normalization is ensured. No stemming or stopword removal is applied because abstractive models operate on raw subword tokens.

4) *Tokenizer-aware truncation*: Since different models allow varying maximum context lengths. During training, we tokenize each MD&A section with the corresponding model tokenizer and truncate to match the model’s effective maximum input length:

- BART / PEGASUS variants: ≈ 512 –768 tokens.
- LED and BigBird–Pegasus: up to ≈ 4096 tokens.
- LongT5: ≈ 1536 tokens.

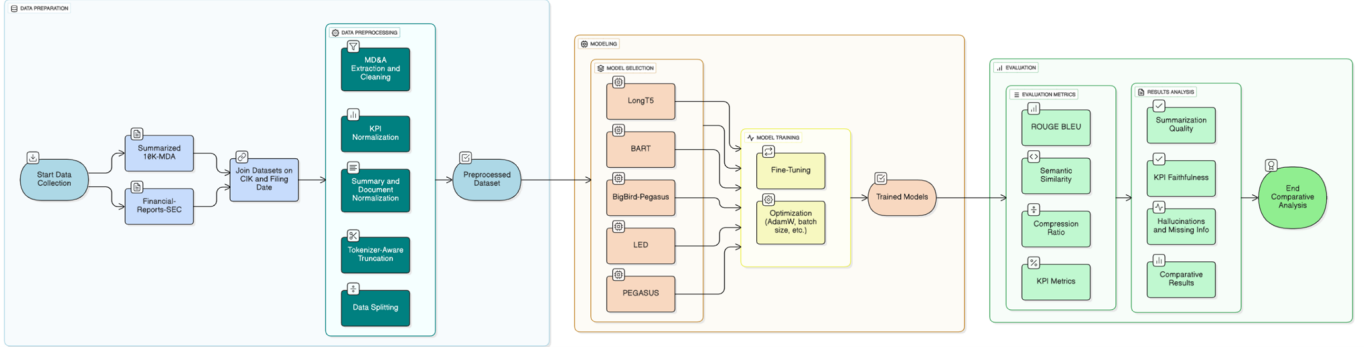


Fig. 1. End-to-end KPI-aware financial summarization pipeline, showing data collection and joining, preprocessing, long-context model fine-tuning, and multi-metric evaluation.

Targets (gold summaries) are truncated to a fixed upper bound (≈ 256 tokens), which is sufficient for the typical summary length observed in Summarized_10K-M&A.

5) *Data Splits*: After joining and cleaning, we randomly split the 153 firm-year instances into:

- Train: 122 MD&A documents ($\approx 80\%$)
- Validation: 15 documents ($\approx 10\%$)
- Test: 16 documents ($\approx 10\%$)

The splits are stratified only by random seed (we do not apply sector-wise splits due to the small size). We keep the same split across all models to enable paired comparisons on a consistent test set.

C. Model Usage

We fine-tune a family of encoder-decoder transformers:

- **BART-base** [9] and a lightly tuned variant **BART-v2** (modified learning rate / batch size and dynamic input length).
- **LED-base** [10], **LED-v2** (dynamic length tuning) and **LED-full** (maximal context utilization).
- **PEGASUS-XSum** [11] and **PEGASUS-v2** (with context-length adaptation).
- **BigBird-Pegasus** [8] as a classic sparse attention long-context baseline.
- **LongT5** [12] as a long-context sequence-to-sequence model.

The training process for all models consists of one epoch of training data processing with AdamW optimization. Because the dataset is small and GPU memory is limited (single 16 GB T4). The training process uses batch size 1 and gradient accumulation and disables mixed-precision for the most memory-intensive models because the dataset size is small and the T4 GPU has 16 GB of memory.

D. Evaluation Metrics

Evaluation is carried out on the held-out test set using:

- **Lexical metrics**: ROUGE-1/2/L and BLEU.
- **Embedding-based semantic similarity**: Cosine similarity between sentence-transformer embeddings of generated vs. gold summaries.

- **Compression ratio**: Summary length divided by MD&A length.
- **KPI metrics**: Precision, recall, and F1 were computed based on overlap between the set of KPIs mentioned in the summary and the normalized document-level KPI set, as well as counts of hallucinated KPIs and missing KPIs per document.

E. Implementation Environment

All experiments were implemented in Python using a hosted Kaggle Notebook environment with a single NVIDIA GPU (≈ 16 GB VRAM) and 12–13 GB of available system RAM. Models were trained with PyTorch and the HuggingFace ecosystem, including transformers for model implementations, datasets for data loading, and evaluate/sacrebleu for metric computation, along with pandas, numpy, and scikit-learn for preprocessing and analysis. Due to the limited GPU memory, we used small batch sizes, gradient checkpointing where possible, and truncated maximum input lengths for some models, which constrained us to short fine-tuning schedules (typically a single epoch). All random seeds, hyperparameters, and code were kept consistent across models to ensure a fair comparison under these resource-constrained conditions.

IV. RESULTS

A. Overall Summarization Quality

Table 1 shows ROUGE-1/2/L, BLEU, KPI-F1, semantic similarity, and compression ratio for all models. At a high level:

- **BART-v2** secured the best ROUGE and BLEU scores among all systems. The model attains ROUGE-1 which is roughly around the mid-0.3 range and ROUGE-L around 0.24, outperforming the vanilla BART and PEGASUS variants. This demonstrates that, on a small but precisely refined dataset, a finely tuned dense-attention model can still achieve performance comparable to that of more complex long-context architectures.
- The ROUGE scores of **LED-full** and **LED-v2** are slightly less than BART-v2 but higher than the original LED configuration, indicating that properly using the available

TABLE I
MAIN RESULTS: MODEL PERFORMANCE COMPARISON

Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	KPI F1	SemSim	Comp. Ratio
bart_v2	0.3417	0.1357	0.2373	0.0639	0.0708	0.5578	0.0279
led_full	0.2793	0.1028	0.2153	0.0050	0.0625	0.4377	0.0141
led_v2	0.2665	0.0712	0.1930	0.0089	0.0625	0.4727	0.0233
bigbird	0.2286	0.0727	0.1615	0.0077	0.0703	0.3100	0.0166
bart	0.2180	0.0659	0.1695	0.0050	0.0625	0.4349	0.0281
pegasus	0.2154	0.0707	0.1568	0.0128	0.0625	0.4743	0.0143
led	0.2085	0.0554	0.1463	0.0032	0.0625	0.3986	0.0127
pegasus_v2	0.2037	0.0673	0.1374	0.0061	0.0625	0.4343	0.0142
longt5	0.0374	0.0004	0.0334	0.0000	0.0625	0.0109	0.1280

long context (via dynamic max-length tuning) results in measurable improvements.

- The ROUGE scores for **BigBird-Pegasus** are moderate, outperforming the weakest baselines but performs low than BART-v2 and LED variants. Its sparse attention is good to deal with lengthy MD&A sections but, since the dataset is relatively small, it limits the observable improvements in lexical overlap.
- **LongT5** did not perform well on ROUGE and BLEU while showing the largest compression ratio. The model tends to produce very short summaries; this is consistent with the high compression-ratio mean in the table I and explains its low lexical scores.
- **KPI-F1** remains low, around 0.06–0.07, for all the models. It’s value slightly exceeded 0.07 even for the best model (BART-v2), which indicates that models failed to reproduce explicit KPI mentions in the reference summaries.
- The **semantic similarity scores** (≈ 0.30 – 0.56) are higher than indicated lexical metrics. BART-v2 got the highest semantic similarity, followed by LED-v2 and LED-full. This indicates that, even though many numeric or fine-grained details are missed, the models capture the overall MD&A with reasonable fidelity.

Overall, the results indicate that **fine-tuned BART-v2** is the strongest general summarizer on this small MD&A dataset, with LED variants leveraging longer contexts but not significantly surpass BART for lexical metrics.

B. KPI Faithfulness, Hallucinations, and Missing Information

Table 2 summarizes KPI-level behavior:

- Each MD&A document in the dataset has on average 3 ground-truth KPIs ($\text{avg_doc_kpis} = 3.0$ for all models).
- Even the best systems mention only about 0.25 KPIs per summary on average (avg_sum_kpis around 0.25 for BigBird and BART-v2), and the number of correctly matched KPIs is even smaller (≈ 0.13 – 0.19). This means that most KPI information is dropped in generated summaries.
- The hallucination rate is relatively low but non-zero: roughly 0.06–0.13 KPI mentions per summary are hallucinated (i.e., appear in the summary but not in the

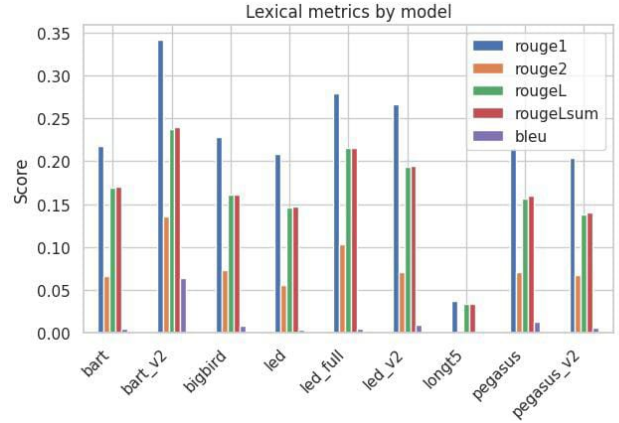


Fig. 2. Lexical overlap of generated summaries with the reference MD&A summaries, measured by ROUGE-1/2/L/Lsum and BLEU on the held-out test set.

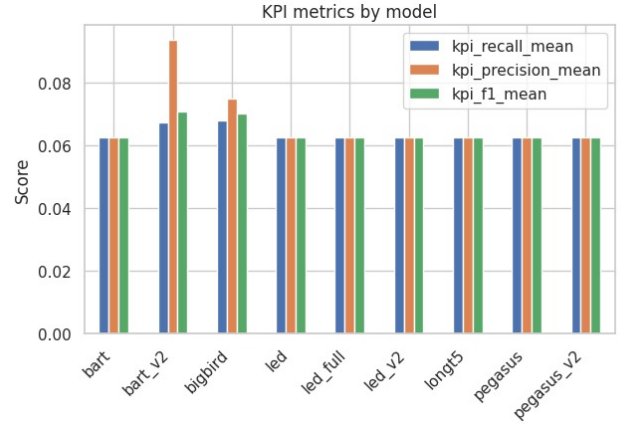


Fig. 3. KPI-level performance of all models on the 10-K MD&A test set.

document-level KPI set). Models thus err more often by omission than by fabrication.

- Consequently, the average number of missing KPIs per document is close to the full 3.0 for most models, slightly lower (≈ 2.8 – 2.9) for BigBird and BART-v2. This confirms that KPI coverage is the main weakness

TABLE II
KPI EVALUATION METRICS ACROSS MODELS (3 KPI CASE)

Model	Avg Doc KPIs	Avg Sum KPIs	Avg KPI Matches	Avg KPI Hallucinated	Avg KPI Missing
bigbird	3.000	0.250	0.188	0.063	2.813
bart_v2	3.000	0.250	0.125	0.125	2.875
pegasus_v2	3.000	0.125	0.063	0.063	2.938
led_full	3.000	0.000	0.000	0.000	3.000
led_v2	3.000	0.125	0.000	0.125	3.000
bart	3.000	0.000	0.000	0.000	3.000
pegasus	3.000	0.063	0.000	0.063	3.000
led	3.000	0.063	0.000	0.063	3.000
longt5	3.000	0.000	0.000	0.000	3.000

across architectures.

These findings justify the focus of the paper: standard summarization objectives and metrics are not sufficient for financial MD&A summarization, because models learn to capture qualitative narrative but fail to reliably transmit the quantitative KPI signals that investors care about.

V. CONCLUSION

In this work, we used a family of longcontext transformer models to study KPI-aware abstractive summarization of 10-K MD&A sections. We created an evaluation benchmark by merging the Summarized 10K MD&A dataset with the financial-reports-sec corpus. Each example includes the complete MD&A document, a human-readable abstractive summary, and a normalized set of document-level financial KPIs. In addition to this benchmark, we established a unified pipeline for training and evaluation that supports both long-context architectures (LED, LongT5, BigBird-Pegasus) and dense-attention models (BART, PEGASUS).

Our tests demonstrate the significant variations in behavior between KPI faithfulness and standard summarization metrics. LED variants make up a competitive second tier, while a tuned BART variant (BART-v2) achieves the best ROUGE and BLEU scores as well as the highest semantic similarity. Although they have access to the entire MD&A section, long-context models like LED and BigBird-Pegasus do not significantly outperform dense models on lexical metrics at this scale. However, KPI-level F1 is still low across all systems, and most summaries leave out most ground-truth KPIs. The modest but non-negligible hallucination rate suggests that existing models are more likely to overlook significant financial signals than to create them.

The limited hardware environment in this study—all training was conducted on a single ≈ 16 GB GPU in a hosted notebook setting—is a practical limitation. Due to this limitation, we had to employ small batch sizes, single-epoch fine-tuning, shorter maximum input lengths, and partial-data training for some long-context models. Therefore, it is important to consider the reported scores as lower-bound estimates of what these architectures could accomplish with larger memory budgets and longer training schedules.

These results point to a significant gap: from the standpoint of financial analysis, models that appear strong under ROUGE/BLEU may still be weak. Our KPI-aware assessment, which is based on the number of matched, hallucinated, and absent KPIs—makes this gap clear and offers MD&A narratives a more domain-relevant perspective on summary quality.

VI. FUTURE WORK

Our future research will focus on expanding the dataset while removing GPU restrictions to test joint training methods between KPI-extraction heads and constrained decoding and reinforcement-learning and preference-based objectives that optimize KPI maintenance. The pipeline shows potential for expansion through the use of larger language models and retrieval-augmented architectures. The results indicate that transformers with extended context capabilities serve as essential components for financial summarization but require KPI-focused training and assessment methods.

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